Testing Plan: Finance & Habit Pattern Finder

Topic 9 – Google Opal

1. Testing Overview

This document describes the test scenarios used to validate the Finance & Habit Pattern Finder workflow.

The testing focuses on transaction input collection, historical memory retrieval, spending analysis, financial recommendations, transaction history preservation, and browser-based reporting.

2. Testing Objectives

- Verify that all transaction inputs are captured correctly.
- Verify that the workflow can handle an initial transaction.
- Verify that previous transaction history can be retrieved.
- Verify that current and historical transactions can be compared.
- Verify spending calculations and category observations.
- Verify that new transactions are appended without removing earlier records.
- Verify that the financial report displays all available transactions.
- Verify that recommendations are based on available data.

3. Test Case 1 – Initial Transaction

Purpose: Verify initial transaction handling when no previous history is available.

Test Input:

Field	Value
Transaction Date	2023-10-27 (date shown in the recorded output)
Amount	$120

Category	Food
Description	Business Lunch

Expected Results:

1. Accept the transaction details.
2. Handle the absence of historical records.
3. Avoid unsupported historical comparisons.
4. Store the first transaction in persistent memory.
5. Display the transaction in the financial report.

Observed Results:

- The report displayed the $120 Food transaction.
- It identified the record as an initial data point.
- The transaction archive displayed the available transaction.
- The output reported that transaction history had been updated.

Result: PASS – Initial transaction processing was demonstrated.

Observation: The recorded output displayed 2023-10-27, so this date is documented as shown rather than assuming a different date.

4. Test Case 2 – Historical Transaction Comparison

Purpose: Verify that the workflow retrieves an earlier transaction and compares it with a new transaction.

Test Input:

Field	Value
Transaction Date	2026-11-07
Amount	$450
Category	Food
Description	Family dinner

Expected Results:

1. Retrieve the previous Food transaction.
2. Compare the new amount with available history.
3. Calculate the combined spending total.
4. Preserve previous transaction records.
5. Generate practical financial recommendations.

Observed Results:

- The report compared the $450 transaction with the previous $120 Food transaction.
- Combined recorded Food spending was calculated as $570.
- It recognized that two data points were insufficient to establish a long-term spending trend.
- The archive displayed both transactions.
- The output reported a successful me